Sahaj Bhadja

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Summary

Software engineer with 2+ years delivering **full-stack**, **cloud**, mobile, and AI products using Python, JavaScript/TypeScript, React, and SQL. Experienced translating ambiguous requirements into tested, production-ready solutions across industries.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Software Engineering, Web Technologies, Database Systems, Analysis of Algorithms

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Architected a multi-tenant AI education platform on Azure SQL and Blob Storage for 500+ users, reducing dashboard latency from 1.2 seconds to 350 milliseconds through query and indexing optimization.
- Built ETL-style integrations using FastAPI, Supabase, SQL Server, and cloud storage, processing 5,000+ monthly jobs with validation, error handling, documentation, and CI/CD-backed releases achieving 99%+ production reliability.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Engineered a React and Python grading platform using multimodal AI and OCR, handling 10,000+ daily submissions with 90%+ consistency while delivering clear workflows and feedback to 150+ instructors.
- Designed secure file ingestion and concurrent batch processing with REST APIs, AWS RDS, and automated delivery, reducing feedback turnaround 60% while maintaining stable performance under peak production workloads.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Engineered cross-platform data-sharing software with automated builds and 20+ unit and integration test suites using Jenkins, PyTest, and Mockito while collaborating with Microsoft and GitHub stakeholders in Agile sprints.
- Optimized complex API integrations using OOP, AWS Glue, and S3, improving response times 30%; led ETL workflow changes that increased pipeline efficiency 68% and reduced technical debt 20%.

Aasma Technology Solutions **India**
Full Stack Developer Intern – VibrantChem *May 2023 – August 2023*
- Developed a MERN resource-management platform with WebSockets, REST APIs, and eight microservices supporting customer, lead, billing, permissions, and notification workflows while synchronizing more than two million records.
- Automated Docker and Jenkins deployments, cutting release time 50%, while Google Analytics insights reduced resource waste 42%, increased lead conversions 15%, and informed iterative improvements to user workflows.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Designed a cross-platform service-booking architecture in Figma, mapped 12+ backend integrations, and presented workflows directly to clients before implementing Flask, Firebase, payment, pricing, notification, and reservation features.
- Delivered real-time mobile and web workflows across 50,000+ records, reducing manual payment handling 70% and maintaining sub-200-millisecond booking responses through API design, testing, and synchronized cloud data.

Skills

Languages: JavaScript, TypeScript, Python, Java, C++, SQL, C#, Bash, HTML, CSS
Frontend & Mobile: React.js, React Native, Next.js, Material UI, Tailwind CSS, responsive design, Figma, accessibility
Backend & Data: FastAPI, Flask, Node.js, REST APIs, WebSockets, PostgreSQL, MySQL, SQL Server, MongoDB, Firebase, ETL
Cloud & Delivery: AWS, Azure, GCP, Docker, Kubernetes, Jenkins, GitHub Actions, Git, CI/CD, PyTest, Mockito, Sentry
Engineering: Agile, Scrum, OOP, data structures, system design, automated testing, code reviews, debugging, monitoring, technical documentation
AI & Tools: Claude, Codex, Cursor, LLM APIs, OCR, RAG, semantic search, model evaluation, Jira, Linux

Projects

Connected Asset Console | *React, TypeScript, Python, FastAPI, WebSockets, PyTest*
- Built and production-compiled an IoT operations prototype with validated telemetry ingestion, deterministic alert rules, latest-state APIs, WebSocket dashboard updates, and automated tests covering threshold classification and state replacement.

PiXMod | Live | *Next.js, AI Models, MongoDB, Stripe*
- Built an AI-powered SaaS platform using Next.js, MongoDB, CLIP, SAM, and Stable Diffusion, processing 1,000+ edited assets while supporting semantic search, subscription billing, and an 18% premium-conversion increase.

HMIS | Live | *React, TypeScript, Healthcare APIs, Sentry*
- Built a healthcare platform with consent-based records access, scheduling, notifications, responsive dashboards, and Sentry monitoring, reducing administrative workload 30% while enabling doctors to retrieve patient histories twice as fast.